Quiz 1 Review: Formulas from Lectures 1–7

Fall 2026 Precalculus. Angles are in radians unless a degree symbol appears, and $k \in \mathbb{Z}$ means k is any integer.

Lecture 1: Coordinates, Lines, and Circles

Idea	Formula
Distance from (x_1, y_1) to (x_2, y_2)	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M = \left(\dfrac{x_1 + x_2}{2},\ \dfrac{y_1 + y_2}{2}\right)$
Circle with center (h, k) and radius r	$(x - h)^2 + (y - k)^2 = r^2$
Slope ($x_1 \neq x_2$)	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$
Point-slope form	$y - y_1 = m(x - x_1)$
Slope-intercept form	$y = mx + b$
General form (includes vertical lines)	$Ax + By + C = 0$, with A, B not both 0
Parallel lines	$m_1 = m_2$
Perpendicular lines	$m_1 m_2 = -1$, so the new slope is $-1/m$
Intersection of $y = m_1 x + b_1$ and $y = m_2 x + b_2$	$x = \dfrac{b_2 - b_1}{m_1 - m_2}$ when $m_1 \neq m_2$
Centroid of a triangle	$G = \left(\dfrac{x_1 + x_2 + x_3}{3},\ \dfrac{y_1 + y_2 + y_3}{3}\right)$
Altitude from a vertex	line through the vertex with slope $-1/m$, where m is the slope of the opposite side

Slope sign: $m > 0$ rises, $m < 0$ falls, $m = 0$ is horizontal, and an undefined slope is vertical. The orthocenter is where the three altitudes meet.

Lecture 2: Functions, Graphs, and Transformations

Parent functions.

Function	$y = x$	$y = x^2$	$y = x^3$	$y = 1/x$	$y = \sqrt{x}$	$y = \lvert x \rvert$
Domain	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R}$	$x \neq 0$	$[0, \infty)$	$\mathbb{R}$
Range	$\mathbb{R}$	$[0, \infty)$	$\mathbb{R}$	$y \neq 0$	$[0, \infty)$	$[0, \infty)$
Symmetry	odd	even	odd	odd	neither	even

Even and odd. f is even if $f(-x) = f(x)$ (symmetric about the y-axis) and odd if $f(-x) = -f(x)$ (symmetric about the origin).

Transformations. A point (a, b) on $y = f(x)$ moves as follows.

New function	New point	Effect
$f(x) + k$	$(a,\ b + k)$	shift up k
$f(x - h)$	$(a + h,\ b)$	shift right h
$-f(x)$	$(a,\ -b)$	reflect across the x-axis
$f(-x)$	$(-a,\ b)$	reflect across the y-axis
$af(x)$	$(a,\ ab)$	vertical scale by a

Quadratics. $y = a(x - h)^2 + k$ has vertex (h, k) and opens up if $a > 0$, down if $a < 0$. For $y = ax^2 + bx + c$, complete the square:

$$h = -\frac{b}{2a}, \qquad k = c - \frac{b^2}{4a} = f(h).$$

Cubics. $y = a(x - h)^3 + k$ has no turning points. **Composition.** $(f \circ g)(x) = f(g(x))$: apply g first, then f.

Inverse functions. f^{-1} exists exactly when f is one-to-one (horizontal line test). Then $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$, and the graph of f^{-1} is the reflection of the graph of f across $y = x$.

Lecture 3: Exponential Functions

Laws of exponents $(b, c > 0)$:

$$b^m b^n = b^{m+n}, \qquad \frac{b^m}{b^n} = b^{m-n}, \qquad (b^m)^n = b^{mn}, \qquad (bc)^n = b^n c^n,$$

$$b^0 = 1, \qquad b^{-m} = \frac{1}{b^m}, \qquad b^{p/q} = \sqrt[q]{b^p}.$$

Graph of $y = b^x$ $(b > 0,\ b \neq 1)$: domain $(-\infty, \infty)$, range $(0, \infty)$, passes through $(0, 1)$, horizontal asymptote $y = 0$. Increasing if $b > 1$, decreasing if $0 < b < 1$. Also $\left(\frac{1}{b}\right)^x = b^{-x}$, the reflection of b^x across the y-axis.

The number e**.** $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828.$

Model	Formula
Compound interest, n times per year	$A(t) = P\left(1 + \frac{r}{n}\right)^{nt}$
Continuous compounding	$A(t) = Pe^{rt}$
Percentage rate per period	$P(t) = P_0(1 + r)^t$: growth if $r > 0$, decay if $-1 < r < 0$
Half-life h	$A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$
Doubling time d	$P(t) = P_0 \cdot 2^{t/d}$

The transformation rules from Lecture 2 apply to $y = b^x$. A vertical shift k moves the horizontal asymptote to $y = k$.

Lecture 4: Inverse and Logarithmic Functions

Finding an inverse. Write $y = f(x)$, solve for x, swap x and y, and state any domain restriction. The domains and ranges swap:

$$\text{dom}(f^{-1}) = \text{range}(f), \qquad \text{range}(f^{-1}) = \text{dom}(f).$$

Definition $(b > 0,\ b \neq 1,\ x > 0)$:

$$\log_b x = y \quad \Longleftrightarrow \quad b^y = x.$$

$\log x$ means $\log_{10} x$ and $\ln x$ means $\log_e x$. Inverse pairs: $\log_b(b^x) = x$ and $b^{\log_b x} = x$, so $\ln(e^x) = x$ and $e^{\ln x} = x$. Special values: $\log_b 1 = 0$ and $\log_b b = 1$.

Rule	Formula
Product	$\log_b(MN) = \log_b M + \log_b N$
Quotient	$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$
Power	$\log_b(M^r) = r \log_b M$
Change of base	$\log_b C = \frac{\ln C}{\ln b} = \frac{\log C}{\log b}$
Exponential equation	$a^x = C \implies x = \frac{\ln C}{\ln a}$
Time in $A = Pe^{rt}$	$t = \frac{\ln(A/P)}{r}$

Solving log equations. Find the domain first (every argument must be positive), combine the logs, then **check every candidate** in the original equation and reject extraneous ones.

Graph of $y = \log_b x$**:** domain $(0, \infty)$, range $(-\infty, \infty)$, x-intercept $(1, 0)$, vertical asymptote $x = 0$. For $y = A\log_b(x - h) + k$, the domain is $x > h$ and the vertical asymptote is $x = h$. The vertical shift k does not move the asymptote.

Applications: $\text{pH} = -\log_{10}[\text{H}^+]$, decibels $L = 10\log_{10}(I/I_0)$, and Richter magnitude $M = \log_{10}(A/A_0)$.

Lecture 5: Angles and Trigonometric Functions

Right triangles. $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$. Special triangles have side ratios $1 : 1 : \sqrt{2}$ (45°-45°-90°) and $1 : \sqrt{3} : 2$ (30°-60°-90°).

Angle measure. $180^\circ = \pi$ radians. To convert degrees to radians, multiply by $\dfrac{\pi}{180}$; to convert radians to degrees, multiply by $\dfrac{180}{\pi}$. Coterminal angles: $\theta + 2k\pi$ (or $\theta + 360^\circ k$).

Arc length and sector area (θ in radians):

$$s = r\theta, \qquad A = \tfrac{1}{2}r^2\theta.$$

Any angle. For a point (x, y) on the terminal side, with $r = \sqrt{x^2 + y^2}$:

$$\sin\theta = \frac{y}{r}, \quad \cos\theta = \frac{x}{r}, \quad \tan\theta = \frac{y}{x}, \qquad \csc\theta = \frac{r}{y}, \quad \sec\theta = \frac{r}{x}, \quad \cot\theta = \frac{x}{y}.$$

On the unit circle, $(\cos\theta, \sin\theta) = (x, y)$.

Special angles.

θ	0	$\pi/6 = 30^\circ$	$\pi/4 = 45^\circ$	$\pi/3 = 60^\circ$	$\pi/2 = 90^\circ$
$\sin\theta$	0	$\dfrac{1}{2}$	$\dfrac{\sqrt{2}}{2}$	$\dfrac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\dfrac{\sqrt{3}}{2}$	$\dfrac{\sqrt{2}}{2}$	$\dfrac{1}{2}$	0
$\tan\theta$	0	$\dfrac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Signs (ASTC: All Students Take Calculus). Quadrant I: all positive. II: sin (and csc) positive. III: tan (and cot) positive. IV: cos (and sec) positive.

Reduction by symmetry to the acute angle α:

Quadrant	α	$\sin\theta$	$\cos\theta$	$\tan\theta$
II	$\pi - \theta$	$\sin\alpha$	$-\cos\alpha$	$-\tan\alpha$
III	$\theta - \pi$	$-\sin\alpha$	$-\cos\alpha$	$\tan\alpha$
IV	$2\pi - \theta$	$-\sin\alpha$	$\cos\alpha$	$-\tan\alpha$

Basic identities.

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta,$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \csc\theta = \frac{1}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta},$$

$$\sin(-\theta) = -\sin\theta, \quad \cos(-\theta) = \cos\theta, \quad \tan(-\theta) = -\tan\theta, \qquad \sin\left(\tfrac{\pi}{2} - \theta\right) = \cos\theta, \quad \cos\left(\tfrac{\pi}{2} - \theta\right) = \sin\theta.$$

Sine and cosine have period 2π and values in $[-1, 1]$; tangent has period π.

Lecture 6: Trigonometric Functions and Their Properties

Triangles. With side a opposite angle A, and so on:

$$\text{Sine law: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \qquad \text{Cosine law: } a^2 = b^2 + c^2 - 2bc\cos A.$$

Graphs of sine and cosine.

	$y = \sin x$	$y = \cos x$
Zeros	$x = k\pi$	$x = \pi/2 + k\pi$
Maximum 1 at	$x = \pi/2 + 2k\pi$	$x = 2k\pi$
Minimum -1 at	$x = 3\pi/2 + 2k\pi$	$x = \pi + 2k\pi$
Symmetry	odd	even

They are shifts of each other: $\cos x = \sin(x + \pi/2)$.

Tangent and reciprocal functions.

Function	Asymptotes (excluded x)	Range	Period	Symmetry
$\tan x$	$\pi/2 + k\pi$	$\mathbb{R}$	π	odd
$\cot x$	$k\pi$	$\mathbb{R}$	π	odd
$\sec x$	$\pi/2 + k\pi$	$(-\infty, -1] \cup [1, \infty)$	2π	even
$\csc x$	$k\pi$	$(-\infty, -1] \cup [1, \infty)$	2π	odd

Sinusoids $y = A\sin(B(x - C)) + D$ and $y = A\cos(B(x - C)) + D$. Factor the input first, for example $2x - \frac{\pi}{2} = 2\left(x - \frac{\pi}{4}\right)$.

Feature	Value
Amplitude	$\lvert A\rvert$
Period	$T = \dfrac{2\pi}{\lvert B\rvert}$
Phase shift	C to the right
Midline and range	$y = D$ and $[D - \lvert A\rvert,\ D + \lvert A\rvert]$

Five-point method ($B > 0$): mark $x = C,\ C + T/4,\ C + T/2,\ C + 3T/4,\ C + T$.

x	C	$C + T/4$	$C + T/2$	$C + 3T/4$	$C + T$
$A\sin(B(x - C)) + D$	D	$D + A$	D	$D - A$	D
$A\cos(B(x - C)) + D$	$D + A$	D	$D - A$	D	$D + A$

Tangent transformed. $y = A\tan(B(x - C)) + D$ has period $\pi/\lvert B\rvert$, range $\mathbb{R}$, no amplitude, and asymptotes where $B(x - C) = \pi/2 + k\pi$.

Lecture 7: Trigonometric Transformations, Equations, and Identities

Point mapping. For $y = Af(B(x - C)) + D$, a point (u, v) on $y = f(x)$ moves to $\left(C + \dfrac{u}{B},\ D + Av\right)$.

Solving equations. Treat the whole inner angle as u, find all solutions for u, then solve for x (scale the k term too). Restrict to an interval only at the end.

Equation	All solutions, $k \in \mathbb{Z}$
$\sin u = \sin \alpha$	$u = \alpha + 2k\pi$ or $u = \pi - \alpha + 2k\pi$
$\cos u = \cos \alpha$	$u = \alpha + 2k\pi$ or $u = -\alpha + 2k\pi$
$\tan u = \tan \alpha$	$u = \alpha + k\pi$

Sum and difference formulas.

	Sum	Difference
Sine	$\sin(a + b) = \sin a \cos b + \cos a \sin b$	$\sin(a - b) = \sin a \cos b - \cos a \sin b$
Cosine	$\cos(a + b) = \cos a \cos b - \sin a \sin b$	$\cos(a - b) = \cos a \cos b + \sin a \sin b$
Tangent	$\tan(a + b) = \dfrac{\tan a + \tan b}{1 - \tan a \tan b}$	$\tan(a - b) = \dfrac{\tan a - \tan b}{1 + \tan a \tan b}$

Double-angle formulas.

$$\sin(2x) = 2 \sin x \cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x, \qquad \tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}.$$

Rewriting squares.

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \qquad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

When an equation has a common factor such as $\sin x$, factor it out instead of dividing by it, or you lose the solutions where that factor is 0.